

Big Data

Chapter 6

NewSQL databases

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you could have best of both worlds?
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Agenda

- New SQL Databases
- Basic properties of traditional RDBMS, NoSQL, and NewSQL
- Transactional properties: ACID, BASE, BASIC
- Architectural assessment of NewSQL systems:
  - MariaDB [ClustrixDB (-> Xpand)]
  - MemSQL (now SingleStoreDB)
  - VoltDB
  - NuoDB
- Comparison of NewSQL systems with respect to feature vectors
- Decision making guide

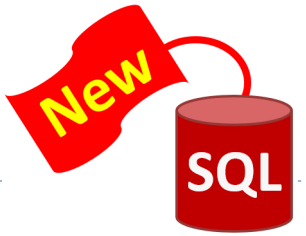
NewSQL Databases

NewSQL databases are a type of relational database system that aims to bridge the gap between traditional relational databases and NoSQL databases.

They aim to offer the scalability and performance of NoSQL databases while maintaining the data consistency and reliability of traditional relational databases.



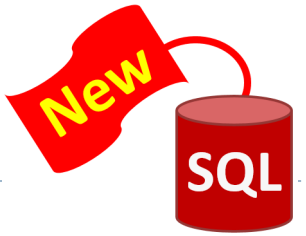
NewSQL Databases



Key characteristics of NewSQL databases:

- Scalability: They can scale horizontally by adding more servers, similar to NoSQL databases. This allows them to handle large datasets and high traffic efficiently.
- ACID compliance: They guarantee Atomicity, Consistency, Isolation, and Durability (ACID) properties, ensuring data integrity and reliability like traditional relational databases.
- SQL querying: They use the familiar SQL language for querying data, making it easier for developers used to relational databases to work with them.
- ACID vs. Speed trade-off: While offering ACID compliance, they might compromise slightly on raw speed compared to some NoSQL databases, but still offer very good performance.

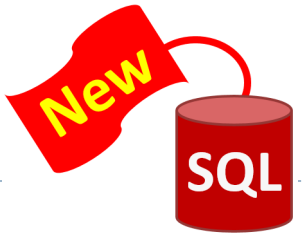
NewSQL Databases



Benefits of NewSQL databases:

- Unified platform: They offer a single platform that combines the strengths of both relational and NoSQL databases, potentially reducing the need for multiple databases.
- Seamless migration: They can be easier to migrate to from existing relational databases due to their SQL compatibility.
- High performance and scalability: They offer good performance and scalability for modern applications.
- ACID guarantees: They ensure data consistency and reliability, important for mission-critical applications.

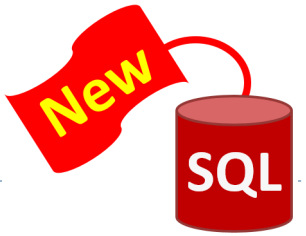
NewSQL Databases



Challenges of NewSQL databases:

- Complexity: They can be more complex to manage and operate compared to traditional relational databases.
- Limited adoption: They are a relatively new technology and might have fewer mature offerings compared to established relational databases.
- Finding balance: Achieving the perfect balance between ACID compliance and speed can be challenging, and the optimal solution depends on specific use cases.

NewSQL Databases



Use cases for NewSQL databases:

- Large-scale e-commerce platforms: Handling high volumes of transactions and user data while ensuring data integrity.
- Real-time applications: Processing and analyzing data streams in real-time with ACID guarantees.
- Financial applications: Requiring high performance, scalability, and data consistency for financial transactions.
- IoT data management: Storing and analyzing sensor data from numerous devices while maintaining data reliability.

NewSQL Databases

Overall, NewSQL databases offer a promising solution for applications requiring both the scalability and performance of NoSQL databases and the data consistency and reliability of traditional relational databases.

However, carefully consider their benefits and challenges, and assess your specific needs to determine if they are the right fit for your project.



NewSQL Databases

How they work?

NewSQL databases achieve their goal of combining scalability and ACID compliance through a mix of innovative technologies and architectural choices:

- Distributed Architecture
- Shared-Nothing vs. Shared-Everything Architectures
- ACID Guarantees
- Concurrency Control
- Querying

NewSQL Databases

How they work?

Distributed Architecture:

Unlike traditional relational databases with a single server, NewSQL databases utilize a distributed architecture, spreading data and workload across multiple servers (nodes) in a cluster. This enables horizontal scaling to handle large datasets and high traffic volumes efficiently.

Shared-Nothing vs. Shared-Everything Architectures:

Some NewSQL databases use a shared-nothing architecture, where each node has its own copy of the data and processes queries independently. This ensures high performance and fault tolerance but can increase storage requirements and complexity.

Other NewSQL databases employ a shared-everything architecture, where data is distributed across nodes but accessed jointly. This can simplify management but might require careful coordination for high performance and consistency.

NewSQL Databases

How they work?

ACID Guarantees:

To ensure data integrity and reliability, NewSQL databases implement mechanisms to achieve ACID properties:

Atomicity: Transactions are treated as indivisible units, ensuring either their complete success or rollback in case of failure.

Consistency: Data transitions from one valid state to another, maintaining data integrity after each transaction.

Isolation: Concurrent transactions remain independent, preventing one from affecting another's results.

Durability: Committed transactions persist even in case of server failures, ensuring data is not lost.

Concurrency Control:

Managing concurrent access to data across multiple nodes is crucial. NewSQL databases employ strategies like optimistic locking or multi-version concurrency control (MVCC) to handle concurrent updates and prevent data conflicts.

NewSQL Databases

How they work?

Querying:

NewSQL databases typically use the familiar SQL language for querying data, making them accessible to developers experienced with traditional relational databases. However, some might have extended functionalities or require adjustments for distributed execution.

Challenges and Trade-offs:

Balancing ACID guarantees with high performance can be challenging. Distributed architectures introduce additional complexity compared to single-server setups.

Finding the optimal solution depends on specific needs. Some NewSQL databases prioritize raw speed, while others emphasize strong consistency guarantees.

NewSQL Databases

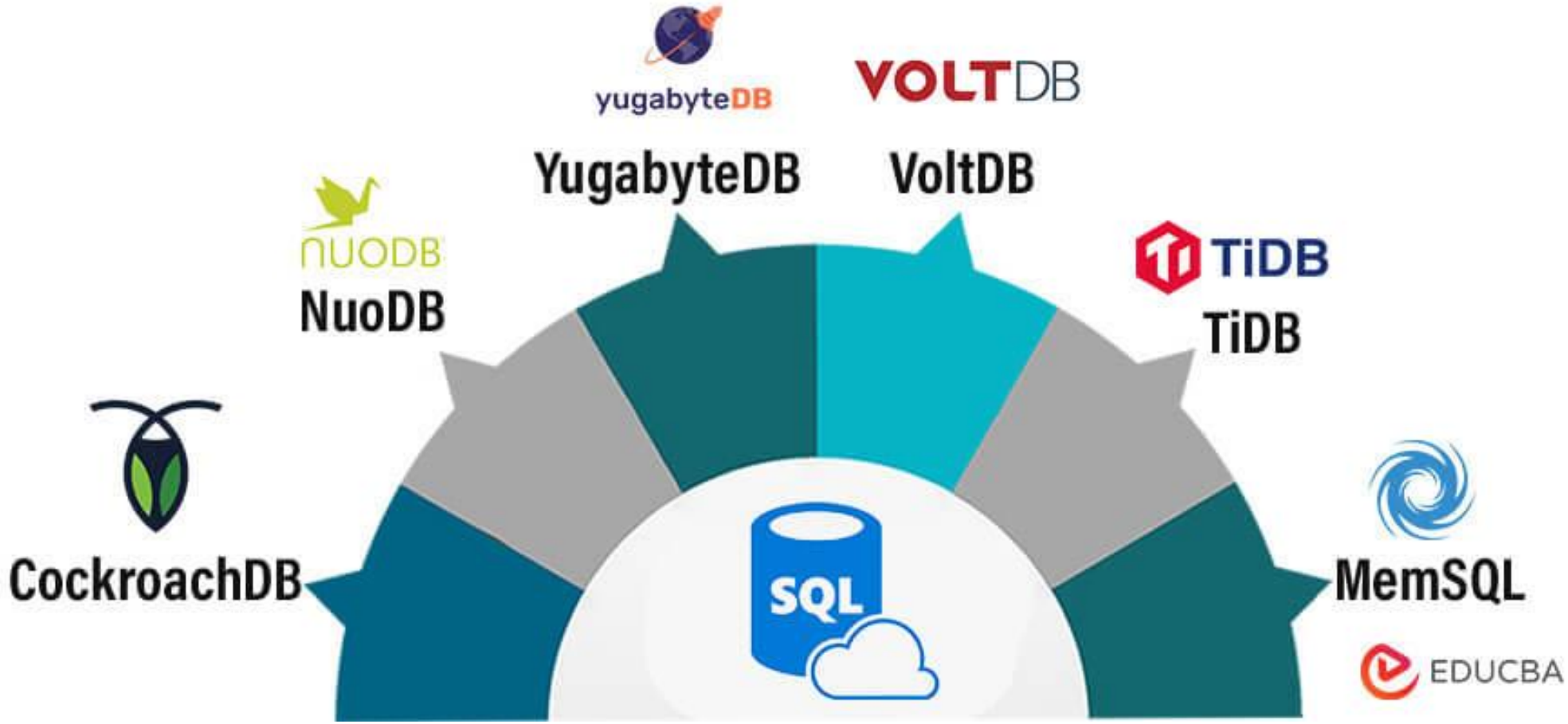
☞ Overall, NewSQL databases represent an exciting evolution in database technology, offering a compelling alternative for applications requiring both scalability and data integrity.

Understanding their inner workings and trade-offs is crucial for evaluating their suitability for our specific project.




SCALEBASE
Tokutek
db Shards
ScaleArc
AKIBAN
ScaleDB
SCHOONER
NewSQL
JustOneDB
nuoDB
SQLFire
TRANSLATTICE
VoltDB
H-Store
GENIEDB
xeround
Clustrix

NewSQL Databases Examples



Most Popular Open-Source NewSQL Databases ChatGPT

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1 **CockroachDB** ☐

- ✅ **Google Spanner-inspired, PostgreSQL-compatible**
- ✅ **Fully open-source (Apache 2.0 license)**
- ✅ **Strong consistency, automatic failover, and geo-distributed transactions**
- ✅ **Ideal for:** Multi-region applications, fintech, e-commerce

🔷 Who Uses It?

- **Bose** – Uses CockroachDB for globally distributed music services.
- **DoorDash** – Uses it for **real-time order processing** at scale.
- **Comcast** – Manages **customer data & authentication** with it.

Most Popular Open-Source NewSQL Databases ChatGPT

2 TiDB

- ✅ **MySQL-compatible, distributed NewSQL database**
- ✅ **Optimized for hybrid transactional & analytical workloads (HTAP)**
- ✅ **Uses Raft for strong consistency**
- ✅ **Ideal for:** Real-time analytics, large-scale OLTP

🔷 Who Uses It?

- **Shopee** – Uses TiDB to handle **millions of daily transactions**.
- **Square (Block, Inc.)** – Uses it for **financial transaction processing**.
- **Meituan** – One of China's largest delivery platforms uses TiDB for **scalable order management**.

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3 YugabyteDB

- ✅ **PostgreSQL-compatible, cloud-native NewSQL database**
- ✅ **Based on Google Spanner's architecture**
- ✅ **Supports ACID transactions with strong consistency**
- ✅ **Ideal for:** High-volume, multi-region applications

🔷 Who Uses It?

- **Walmart** – Uses YugabyteDB to power its **global e-commerce platform**.
- **Kroger** – Uses it for **inventory and supply chain management**.
- **Turvo** – Logistics company using YugabyteDB for **real-time shipment tracking**.

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4 VoltDB ⚡

- ✅ **High-performance, in-memory NewSQL database**
- ✅ **Supports strict ACID compliance & real-time analytics**
- ✅ **Best for:** Telco, finance, and IoT applications

🔷 Who Uses It?

- **Huawei** – Uses VoltDB for **5G network data processing**.
- **Financial institutions** – Use it for **real-time fraud detection**.
- **IoT Platforms** – Uses VoltDB for **high-speed sensor data processing**.

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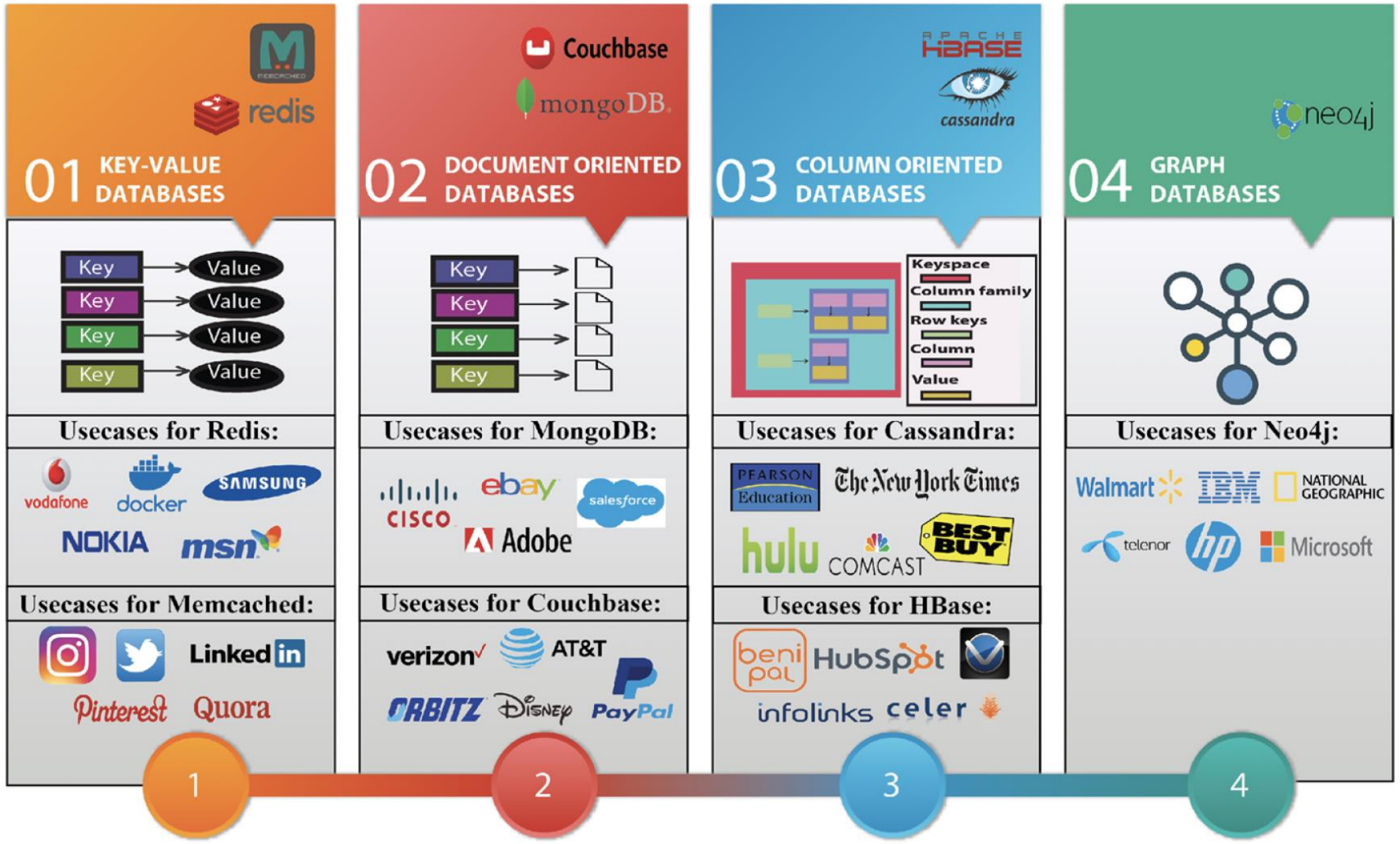
5 NuoDB (Community Edition)

- ✅ **Distributed SQL database for hybrid cloud**
- ✅ **Elastically scales while maintaining SQL support**
- ✅ **Used in banking and enterprise applications**

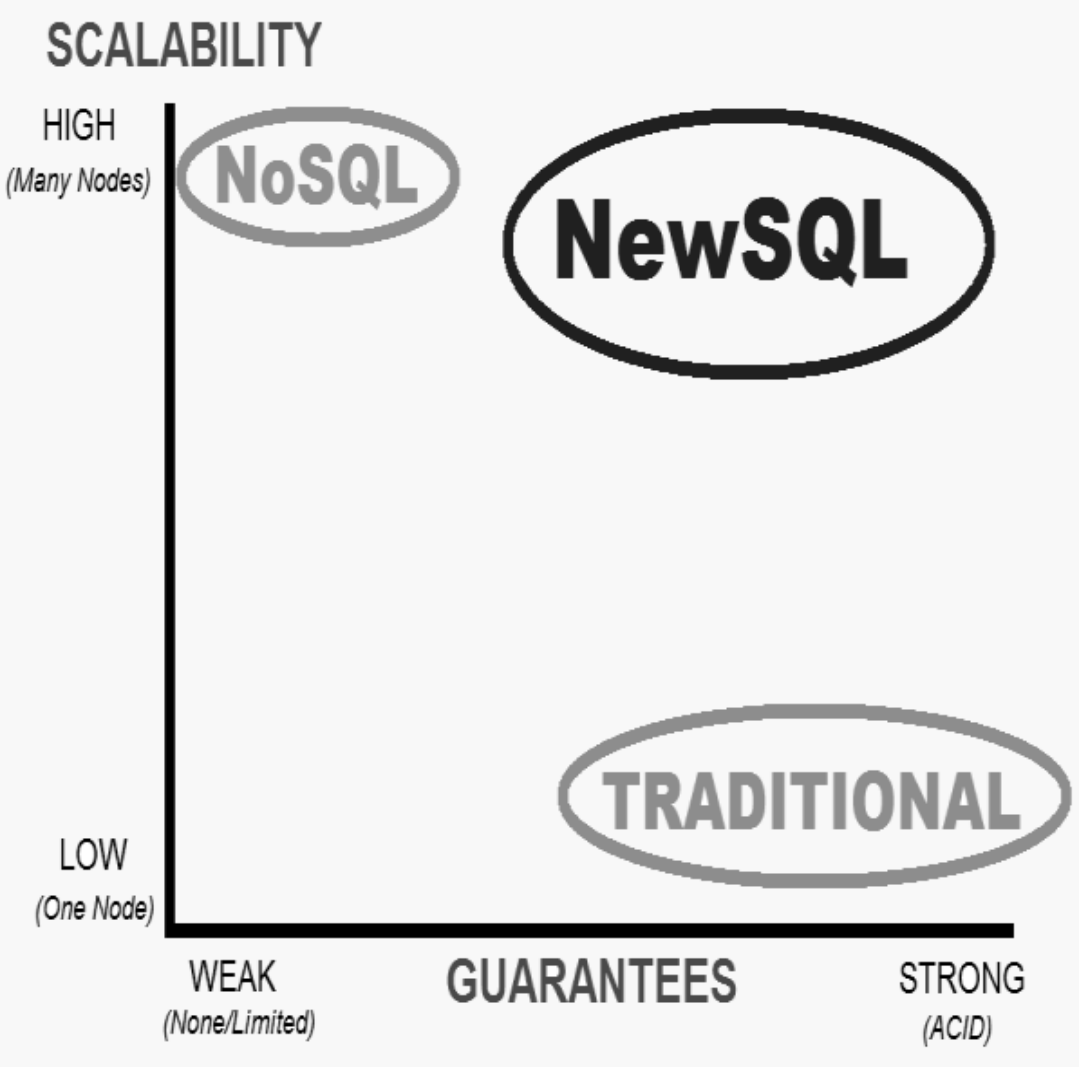
🔷 Who Uses It?

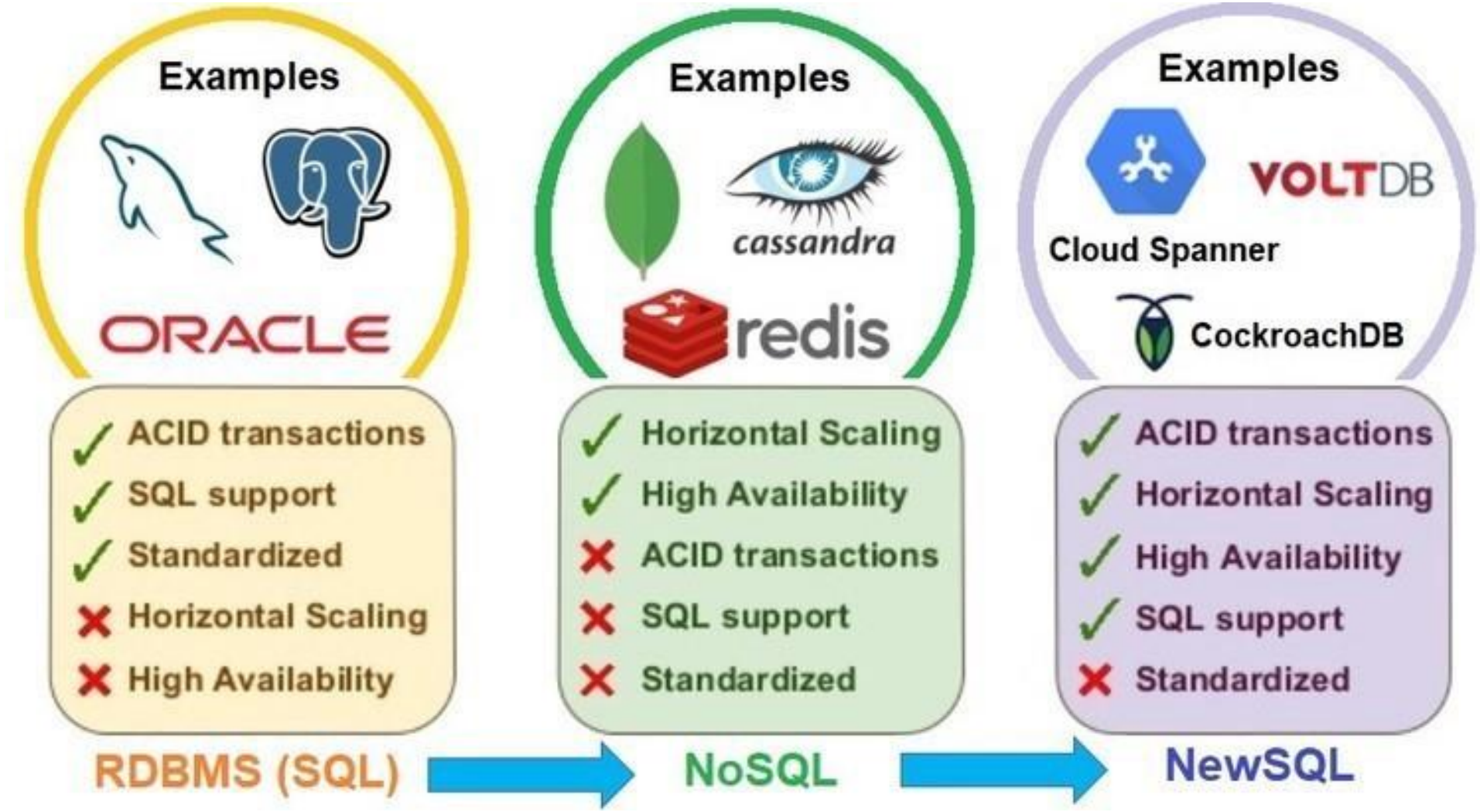
- **Dassault Systèmes** – Uses NuoDB for **enterprise SaaS applications**.
- **Banking sector** – Used for **real-time transaction processing**.

Categories of NoSQL systems



RDBMS vs NoSQL vs NewSQL




Examples
ORACLE
✓ ACID transactions
✓ SQL support
✓ Standardized
✗ Horizontal Scaling
✗ High Availability
RDBMS (SQL)
Examples
cassandra
redis
✓ Horizontal Scaling
✓ High Availability
✗ ACID transactions
✗ SQL support
✗ Standardized
NoSQL
Examples
VOLTDB
Cloud Spanner
CockroachDB
✓ ACID transactions
✓ Horizontal Scaling
✓ High Availability
✓ SQL support
✗ Standardized
NewSQL

RDBMS vs NoSQL vs NewSQL

Functions	RDBMS	NoSQL	NewSQL
SQL	Supported	Not supported	Supported
Machine dependency	Singe machine	Multi- machine/Distributed	Multi- machine/Distributed
DBMS type	Relational	Non- relational	Relational
Schema	Table	Key-value, column store, document store	Both
Storage	On disk + cache	On disk + cache	On disk + cache
Properties support	ACID	CAP through BASE	ACID
Horizontal scalability	Not supported	Supported	Supported
Query Complexity	Low	High	Very High
Security concern	Very high	Low	Low
Big volume	Less performance	Fully supported	Fully supported
OLTP	Not fully supported	Supported	Fully supported
Cloud support	Not fully supported	Supported	Fully supported

Basic properties of traditional RDBMS, NoSQL, and NewSQL

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Properties	SQL	NoSQL	NewSQL
ACID	✓	✗	✓
In-memory DB	✗	✓	✓
Big Data	✗	✓	✓
RDBMS	✓	✗	✓



Transactional properties

- **ACID** is a combination of four properties: Atomicity, Consistency, Isolation, and Durability. These transactional properties of the database guarantee the reliability of transactions.

- **BASE** is an aggregate of three properties namely Basically Available, Soft state, and Eventually consistent. When we have various replicas in a distributed environment, then it is critical to guarantee the consistency among the replicas. Therefore, consistency is achieved eventually.

- **BASIC** is an acronym of Basic Availability, Scalability, and Instant Consistency.

Transactional properties - BASIC

- Basic Availability, Scalability, and Instant Consistency

- Basic Availability confirms that the system always responds to read/write queries.
- Scalability allows for adding more resources if the workload is increased.
- Instant Consistency ensures that if a write and read operation is carried out consecutively, then the result returned by a read operation must be the same as written by a write operation.

Architectural assessment of NewSQL systems

- MariaDB Xpand includes scalability and robustness
- MemSQL provides high performance, horizontal scalability, and relational data model
  - On October 27, 2020, MemSQL rebranded to SingleStore
- VoltDB is an in-memory NewSQL system aimed for high data velocity applications
- NuoDB provides the ACID guarantee.

Feature vectors

- Various properties and aspects that have a significant impact on the performance of a database system

	Components of feature vector
Transactional properties	Atomicity, consistency, isolation, and durability
Fault tolerance	Failure handling, recovery, high availability, support for replication, replication method, replication type, and replica synchronization
Data storage	Data model, data storage, in-memory support, partitioning, and support for secondary indexes
Data handling	Data expiration control, data compression, triggers, support for MapReduce, concurrency control, analytics support, database as a service, and rack-locality awareness

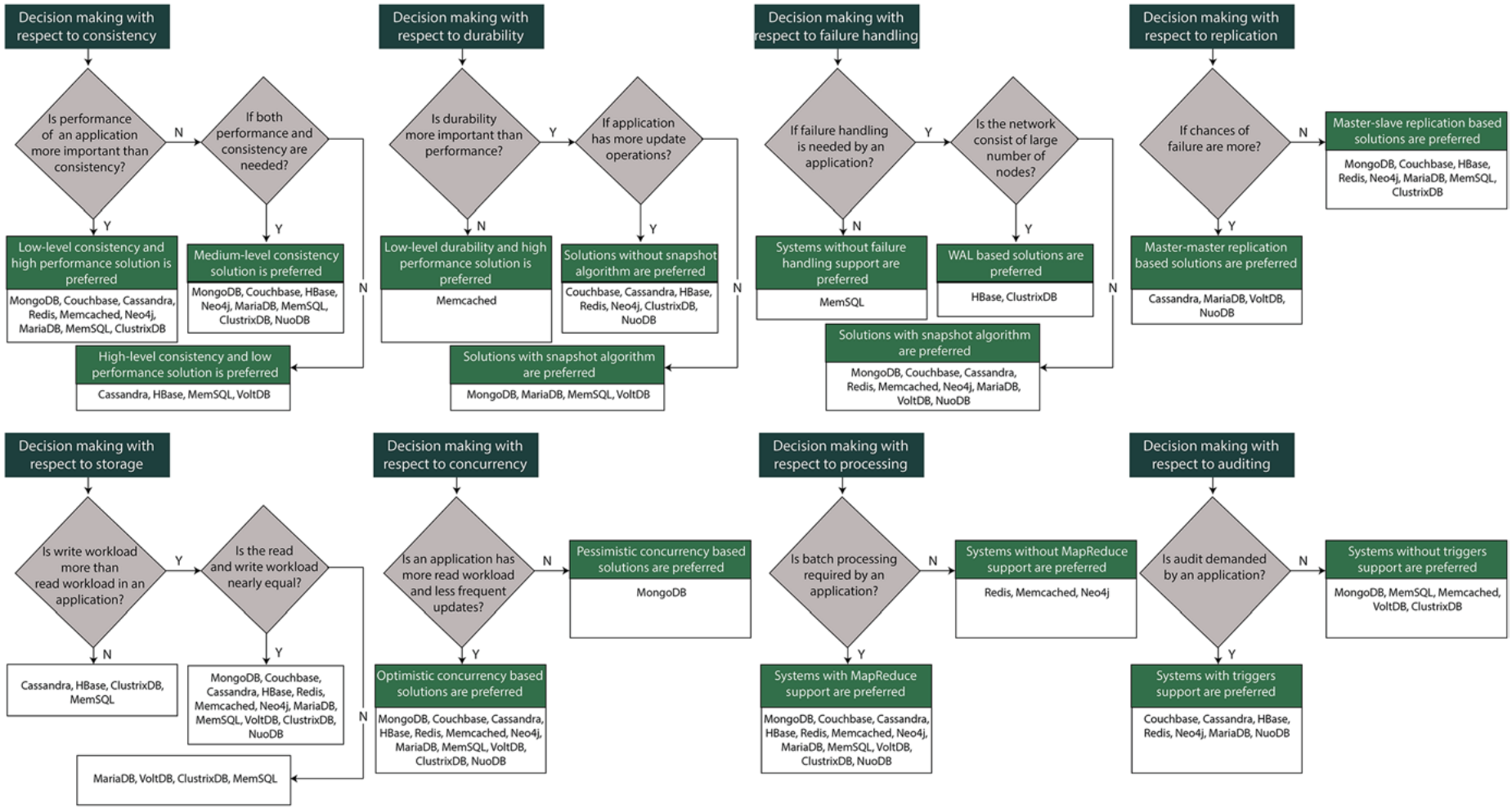
Comparison of NewSQL systems with respect to feature vectors

FVs/NewSQL	MariaDB	MemSQL	VoltDB	Clustrix	NuoDB
Transactional properties	Atomic writes; Consistent snapshots and eventual consistent in master/slave replication; Read committed, serializable, and repeatable read isolation levels; Group commit to binary log and sync data to the disk	Two-phase locking; Strong, Immediate, and eventual consistency; Read committed; Durability using logging on disk and disk based column stores	100% rollback atomicity; Strong consistency; Serializable isolation level; Durability using snapshots, command logging	Two-phase locking, Immediate consistency; Read committed, repeatable read, and Serializable isolation level; Write-ahead logging (WAL)	Roll back atomicity; Strong consistency; Consistent read, write committed, read committed, repeatable read, and serializable; Journaling and safe tunable commit protocol
Fault tolerance	Heart beat period monitoring; Recovery modes exists from 1 to 6&; Multi master and master slave replication; Synchronous, semi synchronous, and asynchronous replication; Synchronization of binary log files	Cluster heart beats; Snapshot restoration; Availability groups; Master slave replication; Asynchronous replication; Exchange of snapshot and log files	Heart beat messages and automated snapshots; Snapshots and command log recovery; Peer-to-peer, master-to-master, active-active replication; Synchronous and asynchronous replication; Automatic synchronization	Heart beat messages and auto-healing; Log and peer-to-peer failure handling; Group membership and quorum protocol; Master and slave replication; Synchronous replication; replica synchronization using WAL	Ping timeouts and Broker quorum; Journal recovery; availability using storage managers; Peer to peer replication; Asynchronous replication; Automatic synchronization of Storage Managers (SMs)
Data storage	XtraDB and various other storage engines; In-memory tables and memory storage engine features; Range partitioning	Both row-based and column based store; In-Memory Rowstore, Flash, SSD, or Disk-based Columnstore; In-memory row store; Hash Partitioning	In memory and disk storage; In-memory logs; Automatic partitioning	In-memory tables and solid state disk (SSD); In memory real time analytics; Consistent hashing	Persistent storage through storage manager (SM): key-value and HDFS and cache; Cache for storing UDFs; Table partitioning to storage groups

Comparison of NewSQL systems with respect to feature vectors

FVs/NewSQL	MariaDB	MemSQL	VoltDB	Clustrix	NuoDB
Data handling	Yes, expiration control of binary logs; Page compression using zlib, lz4, lzo, lzma, bzip2 and snappy compression algorithm; MVCC using Aria storage engine ; Batch, real time and algorithmic analytics; Cross data center replication	Query and connection timeouts; Storage in SSD or disk in compressed format; MVCC, lock free storage engine; Real time analytics support; Cross datacenter replication	Expiration timeout period set for connections; Snappy compression; Concurrency through single-threaded pipeline scheduling; Real time analytics support; Cross datacenter replication	Compressed backup; MVCC, 2PL and Paxos protocol; Real time analytics support; Cross datacenter replication	Expiration timeout period is set for connections, MVCC, Hybrid transactional and analytical processing; Cross datacenter replication

Decision making guide



Source: Chaudhry, N., Yousaf, M.M. Architectural assessment of NoSQL and NewSQL systems. *Distrib Parallel Databases* **38,** 881–926 (2020). https://doi.org/10.1007/s10619-020-07310-1

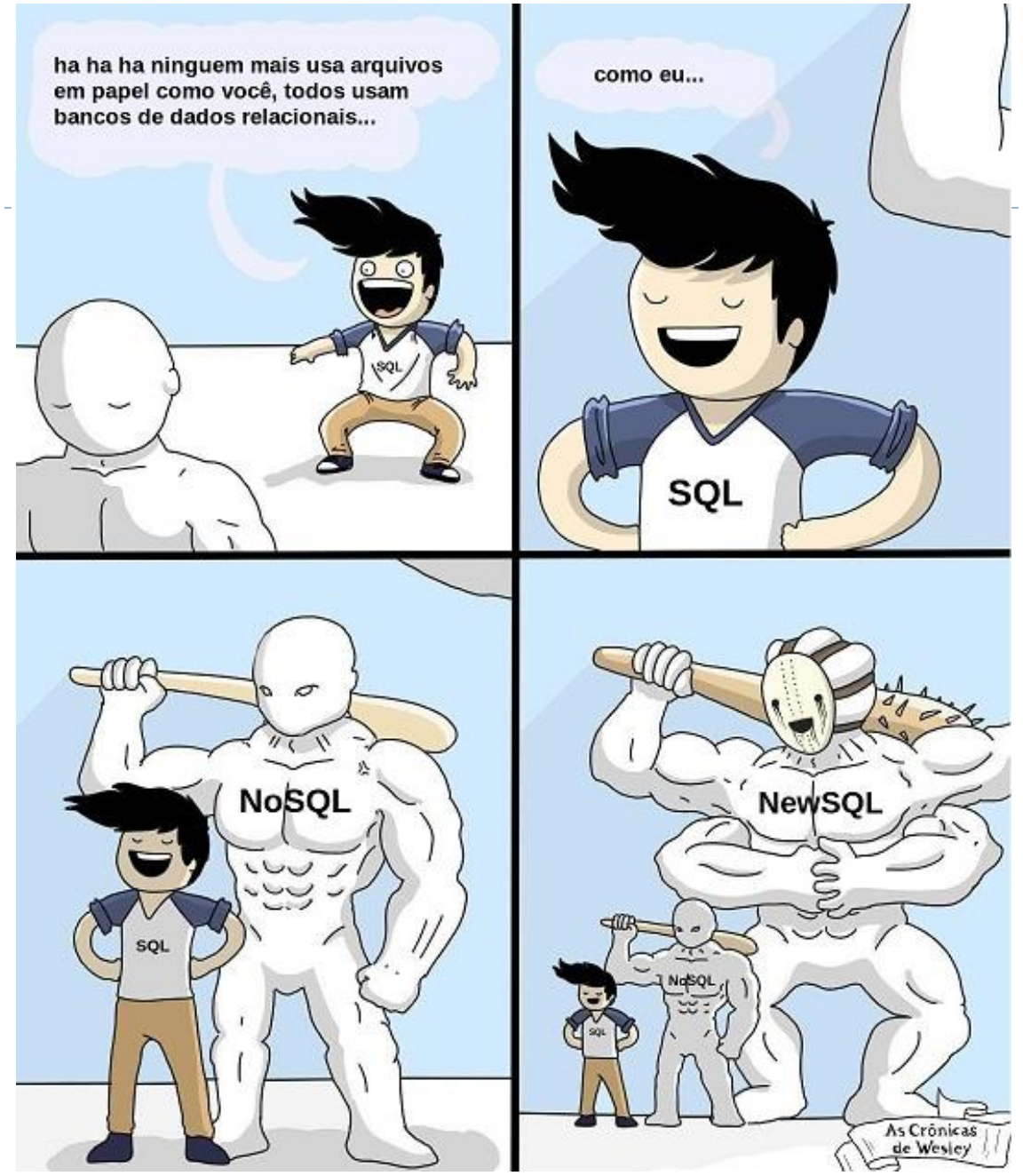
Conclusions

- To choose a right NoSQL or NewSQL system for an application, one must focus on architectural features relevant to application requirements.

- In recent times, many studies have performed architectural assessment of NoSQL and NewSQL systems but these lack detailed comparison of architecture.

Summary

- NuoDB provides a stronger consistency level. If an application has serious consistency concerns then MemSQL, VoltDB, and NuoDB would be a feasible choice.
- However, if speed and performance of application is of concern then MariaDB Xpand should be considered.
- VoltDB can be selected to provide strict isolation at the expense of performance.
- Triggers help to audit and validate data based on business requirements. Triggers can cause performance issues if not used carefully. If the application has strict security requirements then triggers can be a beneficial feature to use. MemSQL, and VoltDB do not have a support for triggers. All other data management systems involved in comparative analysis have a support for triggers.


ha ha ha ninguem mais usa arquivos em papel como você, todos usam bancos de dados relacionais...
SQL
como eu...
SQL
NoSQL
SQL
NewSQL
NoSQL
SQL
As Crônicas de Wesley


ANY
QUESTIONS?

T.P.C. (Homework) To Read

- Carvalho, Inês, Filipe Sá, and Jorge Bernardino. 2023. "Performance Evaluation of NoSQL Document Databases: Couchbase, CouchDB, and MongoDB" *Algorithms* 16, no. 2: 78. https://doi.org/10.3390/a16020078

- Monteiro, Jéssica, Filipe Sá, and Jorge Bernardino. 2023. "Experimental Evaluation of Graph Databases: JanusGraph, Nebula Graph, Neo4j, and TigerGraph" Applied Sciences 13, no. 9: 5770. https://doi.org/10.3390/app13095770

- Pina, Eduardo, Filipe Sá, and Jorge Bernardino. 2023. "NewSQL Databases Assessment: CockroachDB, MariaDB Xpand, and VoltDB" *Future Internet* 15, no. 1: 10. https://doi.org/10.3390/fi15010010